LITERATURE REVIEW

An equilibrium model for spot and forward prices of commodities

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ANALYTICAL ABSTRACT

The paper builds a two-date partial equilibrium for a storable commodity. A representative producer decides how much output to withhold in inventory and how many forward contracts to sell; a representative financial investor, who also trades an exogenously priced stock index, decides how many forwards to buy. Both have exponential utility, and consumer demand is linear with a random shift at the horizon. Existence of an equilibrium forward price is proved by convex duality and sup-convolution for a general set of attainable trading gains. With Lévy-driven demand and stock index, the investor's problem reduces to a minimal entropy martingale measure computation, and the equilibrium is characterised by semi-explicit equations. In Brownian and jump-diffusion examples, numerical comparative statics show that the spot price rises with producer risk aversion and falls with investor risk aversion, and that correlation between demand and the stock index lowers the investor's effective risk aversion, thereby raising spot prices and lowering forward premia. The model is never confronted with data. This review reproduces the Brownian case and shows that the closed-form storage rule holds only where the producer's problem is jointly concave.

1 Introduction

1.1 Paper identification and classification

The paper is a theoretical contribution to commodity pricing with a numerical illustration and no empirical estimation. Its field is financial economics and its subfield is the equilibrium relation between spot prices, forward prices and inventories, treated with the tools of mathematical finance (convex duality, entropic risk measures, Lévy processes). The version read in full is arXiv version 3, in the HTML rendering of the arXiv source, which contains the main text, the proofs and the appendix. The graphical figures were not available in that rendering as images, so the reading of Figures 5.1 to 5.7 relies on their captions and on the authors' discussion, supplemented by the reviewer's own re-computation of the Brownian case reported below.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

It asks how the spot price, the forward price, the inventory policy and the volume of forward trading are jointly determined when risk-averse producers hedge in the forward market against a risk-averse financial investor who can also trade the stock market, and whether participation of the investor and correlation between commodity demand and equities move spot prices and forward premia.

WHY DOES THE QUESTION MATTER?

Since the early 2000s institutional money flowed into commodity futures and commodity returns became correlated with equity returns. Whether spot prices were pushed by this financialisation or by fundamentals is a policy and an asset allocation question. Few structural models contain, in the same optimisation, the producer's storage choice, the forward market and the stock market.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

The classical theories of storage and of normal backwardation relate forward prices to carrying costs, convenience yields and hedging pressure. Equilibrium versions exist with mean-variance preferences or risk-neutral agents, usually with investors confined to the commodity market. What is missing is a model with non-mean-variance preferences, a second traded asset correlated with the commodity shock, and non-Gaussian shocks, in which the spot price is endogenous through storage.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes a static two-agent model with CARA preferences and a linear demand curve shifted by a random variable at the horizon. It proves existence and, under differentiability, uniqueness of the equilibrium forward price, derives semi-explicit conditions under Lévy dynamics, and studies two

parametric examples. It finds that the spot price is increasing in producer risk aversion, decreasing in investor risk aversion, and increasing in squared correlation with the stock index, that forward premia move in the opposite direction, and that the presence of a forward market raises the spot price while lowering the expected future price when output is uneven across dates.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks how the spot price P_0, the forward price F, the storage choice α and the forward volume b are jointly determined at equilibrium when a representative producer with exponential utility, who may store part of output and sell forwards, faces a representative investor with exponential utility, who may trade forwards and a correlated stock index, and how these quantities depend on the two risk aversions, on the correlation between the demand shift and the stock index, on jump risk and on the asymmetry of production across dates.

2.2 Hypotheses

The paper does not state numbered hypotheses; the following are the testable propositions that it derives and discusses (all labelled Paper).

- The equilibrium spot price is increasing in the producers' risk aversion γ_p and decreasing in the investors' risk aversion γ_s.
- Because the investor's effective risk aversion is $\bar{\gamma}_s = \gamma_s(1-\rho^2)$, more correlation (in absolute value) between demand and the stock index acts like a lower γ_s: higher spot price, higher forward volume, lower forward premium.
- A forward market raises the spot price and lowers the expected future spot price when future production is scarce relative to current production, and so stabilises prices.
- The convenience yield is increasing in both risk aversions but not monotone in ρ, because a lower effective γ_s and a higher spot price move it in opposite directions.

2.3 Existing literature and research gap

PAPER The authors position the model against Stoll (1979), Anderson and Danthine (1983) and Hirshleifer (1988, 1989), who use mean-variance preferences; Routledge, Seppi and Spatt (2000) and Pirrong (2011), where agents are risk neutral and forward prices equal expected spot prices; Baker (2014) and Ekeland, Lautier and Villeneuve (2014), with mean-variance investors; and Acharya, Lochstoer and Ramadorai (2013) and Gorton, Hayashi and Rouwenhorst (2012), where agents trade only in the commodity market. The claimed gap is the absence of an exogenous stock market in the investor's choice set, of general risk aversion and of rich factor dynamics.

CRITIQUE The gap is real but narrower than presented. With CARA utility and Gaussian shocks the exponential and mean-variance problems coincide, so the Brownian example is a mean-variance model with a hedgeable component. Novelty beyond the earlier mean-variance work therefore lies in the existence theorem for general trading sets and in the jump example.

2.4 Claimed contribution

CRITIQUE The paper claims (i) an existence theorem, (ii) semi-explicit formulas, and (iii) economic insights on financialisation. Claim (i) is demonstrated, with one reservation on concavity discussed in the critical evaluation. Claim (ii) is demonstrated, though one printed term is dimensionally inconsistent as rendered. Claim (iii) is overstated: the insights are properties of a static, price-setting-producer model with no empirical test, and they depend on a modelling choice (producers internalise their effect on the spot price) that is not discussed as a driver of the main result.

3 Theoretical Framework and Mechanism

3.1 Agents, timing and spot market

PAPER There are two dates, 0 and T. Producers output the deterministic quantities π_0 and π_T. They sell $\pi_0 - \alpha$ at date 0, store $\alpha \in [0, \pi_0]$, of which a fraction ε is lost, and hold h_p forward contracts struck at F. Terminal wealth is

$$\bar{w}(\alpha, h_p) = P_0(\pi_0 - \alpha)(1 + R) + P_T\big(\pi_T + \alpha(1 - \varepsilon)\big) + h_p\,(P_T - F). \tag{1}$$

P_0, P_T	Spot prices at dates 0 and T.
R	Discretely compounded interest rate: date-0 revenue is carried to T.
$\varepsilon \in [0, 1]$	Storage cost as a fraction of stored units.
h_p	Forward position; negative means short (hedging).
F	Forward price for delivery at T, determined at equilibrium.

Equation (1) says that the producer receives interest-bearing spot revenue today, sells at the random price P_T both next-period output and what was stored, and gains or loses on the forward position. Preferences are exponential,

$$\mathbb{U}_p(v) = -\frac{1}{\gamma_p} \log \mathbb{E}\big[e^{-\gamma_p v}\big], \qquad \Pi_p = \sup_{\alpha \in [0, \pi_0],\, h_p \in \mathbb{R}} \mathbb{U}_p\big(\bar{w}(\alpha, h_p)\big). \tag{2}$$

Consumers' demand is linear and shifts randomly at T,

$$\psi_0(x) = \mu - mx, \qquad \psi_T(x) = \psi_0(x) + X, \qquad \phi_0(y) = \frac{\mu - y}{m}, \quad \phi_T(y) = \frac{\mu + X - y}{m}. \tag{3}$$

Market clearing at both dates gives the spot prices as functions of storage:

$$P_0 = \phi_0(\pi_0 - \alpha) = \phi_0(\pi_0) + \frac{\alpha}{m}, \qquad P_T = \phi_0(\pi_T) - \frac{\alpha(1-\varepsilon)}{m} + \frac{X}{m}. \tag{4}$$

μ, m Demand intercept and slope; $1/m$ is the price response to a unit of quantity.

X Random shift of the demand curve at T, a random variable with exponential moments.

The key economic mechanism is in (4). A unit stored raises P_0 by $1/m$ and lowers the expected P_T by $(1-\varepsilon)/m$: producers act as price setters and trade off current revenue, the price effect on the units they sell, and exposure to P_T. Storing more raises the spot price, but also raises the quantity $\pi_T + \alpha(1-\varepsilon)$ exposed to demand risk.

3.2 The investor and the equilibrium

PAPER The investor takes a forward position h_s and a trading gain $G \in \mathcal{G}$, a convex set of zero-cost attainable outcomes containing 0:

$$\bar{w}(G, h_s) = h_s(P_T - F) + G, \qquad \Pi_s = \sup_{h_s \in \mathbb{R},\, G \in \mathcal{G}} \mathbb{U}_s\big(h_s(P_T - F) + G\big). \tag{5}$$

$\mathcal{G}$ Set of discounted gains from trading the stock index, for example $\{\int_0^T \theta_u \, dS_u : \theta \in \Theta\}$.

γ_s Investor risk aversion, with $\mathbb{U}_s$ defined as in (2).

An equilibrium is a triple $(\hat{\alpha}, \hat{h}, \hat{F})$ with market clearing $\hat{h} = h_p(\hat{F}) = -h_s(\hat{F})$, and optimality of $(\hat{\alpha}, \hat{h})$ for the producer and of $\hat{h}$ for the investor. Because $\mathbb{U}$ is the negative of an entropic risk measure, both problems minimise an entropic risk.

3.3 Existence of the equilibrium

PAPER Write the cumulant generating function of X as $\kappa_X(u) = \log \mathbb{E}[e^{uX}]$. Substituting (4) in (1) gives a quadratic part and a part linear in X,

$$\bar{w} = q(\alpha, h_p) + \ell(\alpha, h_p) X, \qquad \ell(\alpha, h_p) = \frac{\alpha(1-\varepsilon) + h_p + \pi_T}{m}, \tag{6}$$

and translation invariance of exponential utility yields

$$\mathbb{U}_p(\bar{w}) = q(\alpha, h_p) - \frac{1}{\gamma_p} \kappa_X\big(-\gamma_p \, \ell(\alpha, h_p)\big). \tag{7}$$

Since κ_X is convex (Hölder's inequality) and lower semicontinuous (Fatou), the producer's objective is upper semicontinuous and, under condition (COE), coercive in h_p, so a maximiser exists for every F (Proposition 3.3). For the investor, with $u_s(\alpha, h_s) := \sup_{G \in \mathcal{G}} \mathbb{U}_s(h_s P_T + G)$, the dual bound from the

entropy inequality gives, for any $\mathbb{Q} \in \mathcal{M}_{\mathcal{G}}$ with finite relative entropy $\mathcal{H}(\mathbb{Q}|\mathbb{P})$,

$$u_s(\alpha, h_s) \le h_s\, \mathbb{E}_{\mathbb{Q}}[P_T] + \frac{1}{\gamma_s}\mathcal{H}(\mathbb{Q}|\mathbb{P}), \tag{8}$$

so that $F := \mathbb{E}_{\mathbb{Q}}[P_T]$ makes the investor's value finite (Proposition 3.7). Define the sup-convolution of the two agents' value functions,

$$u(\alpha, h) = \sup_{h_p + h_s = h} \{u_p(\alpha, h_p) + u_s(\alpha, h_s)\}, \qquad u^*(\alpha, F) = u_p^*(\alpha, F) + u_s^*(\alpha, F). \tag{9}$$

The equilibrium forward price at fixed α is any element of the supergradient $\partial u(\alpha, 0)$: at $h = 0$, $F \in \partial u(\alpha, 0)$ is exactly the condition that the same price supports both agents' optimal positions with $h_p + h_s = 0$. The supergradient is non-empty because 0 lies in the interior of the domain of $u(\alpha, \cdot)$ (Rockafellar, Theorem 23.4). Theorem 3.9 then lets $\alpha_n \to \alpha$ along a maximising sequence and shows in three steps that $F(\alpha_n)$ and h_n are bounded, that the conjugates converge, and that the limit price equals $F(\alpha)$. Uniqueness of $\hat{F}$ holds when $u_p(\alpha, \cdot)$ and $u_s(\alpha, \cdot)$ are differentiable (Remark 3.10).

CRITIQUE The proof of Proposition 3.3 states that the objective is strictly concave in α and concave in h_p. That is concavity in each variable separately. Existence needs only upper semicontinuity, coercivity and compactness of the α range, so the theorem stands. Joint concavity, which governs whether first-order conditions identify the maximiser, is a different property and is examined in the critical evaluation.

3.4 Lévy dynamics and minimal entropy

PAPER With a d-dimensional Lévy process Z of triplet (b, c, ν) and cumulant generating function $\kappa(u) = \langle u, b\rangle + \frac{1}{2}\langle u, cu\rangle + \int (e^{\langle u, x\rangle} - 1 - \langle u, x\rangle)\nu(dx)$, the stock index and the demand shift are

$$S_t = S_0 e^{Y_t}, \quad Y_t = \langle u_1, Z_t\rangle, \qquad X = \langle u_2, Z_T\rangle, \qquad \kappa_X(v) = T\kappa(vu_2). \tag{10}$$

For a given forward position h_s the investor's problem is split by a change of measure,

$$\frac{d\mathbb{P}^s}{d\mathbb{P}} = \frac{\exp(-\gamma_s h_s X/m)}{\mathbb{E}[\exp(-\gamma_s h_s X/m)]}, \qquad \kappa^s(v) = \kappa(v + \xi) - \kappa(\xi), \quad \xi = -\frac{\gamma_s h_s}{m}u_2. \tag{11}$$

Under $\mathbb{P}^s$ the investor faces a pure stock-market problem, solved by the minimal entropy martingale measure, which for exponential Lévy models is the Esscher measure $d\mathbb{P}^*/d\mathbb{P}^s = e^{\eta^* \tilde{Y}_T}/\mathbb{E}^s[e^{\eta^* \tilde{Y}_T}]$ where η^* solves $\partial_v \tilde{\kappa}_1^s(v)|_{v=\eta^*} = 0$ (condition (FE)), and the value is

$$\sup_{\theta \in \Theta}\left\{-\frac{1}{\gamma_s}\log \mathbb{E}^s\left[e^{-\gamma_s G_T(\theta)}\right]\right\} = -\frac{T}{\gamma_s}\tilde{\kappa}_1^s(\eta^*). \tag{12}$$

Collecting terms, the investor's reduced problem is

$$\Pi_s = \sup_{h_s}\left\{-\frac{T}{\gamma_s}\left(\tilde{\kappa}_1^s(\eta^*) + \kappa_2\left(-\frac{\gamma_s h_s}{m}\right)\right) + h_s\left(\phi_0(\pi_T) - \frac{\alpha(1-\varepsilon)}{m} - F\right)\right\}. \tag{13}$$

The first term is the entropic cost of stock-market investing after the forward position has tilted the measure; the second is the cumulant penalty of the unhedged demand risk; the last is the expected excess return $\mathbb{E}[P_T] - F$ on the forward. The hedge with the stock index enters through $\mathbb{P}^s$: when Y and X are dependent, the tilt changes the drift of Y and hence η^*.

3.5 Brownian example

PAPER Take $Y_t = b_1 t + \sigma_1 W_t^1$, $X_t = \sigma_2 W_t^2$, $\operatorname{corr}(W^1, W^2) = \rho$. Then $\mathbb{E}[P_T] = \phi_0(\pi_T) - \alpha(1-\varepsilon)/m$ and $\operatorname{Var}[P_T] = \sigma_2^2 T/m^2 =: V$. The producer's problem is the quadratic programme

$$\Pi_p = \max_{h_p, \alpha} \{ d_1 \alpha^2 + d_2 \alpha + d_3 \alpha h_p + d_4 h_p^2 + d_5 h_p + d_6 \}, \tag{14}$$

with, from (A.5) of the paper,

$$\begin{aligned} d_1 &= -\Big(\frac{1 + R + (1-\varepsilon)^2}{m} + \frac{\gamma_p (1-\varepsilon)^2}{2} V \Big), & d_3 &= -\frac{1-\varepsilon}{m} - \gamma_p (1-\varepsilon) V, \\ d_4 &= -\frac{\gamma_p}{2} V, & d_5 &= -\Big(F - \frac{\mu - \pi_T}{m} \Big) - \gamma_p \pi_T V, \end{aligned} \tag{15}$$

and $d_2 = \big[2(1+R)\pi_0 - 2(1-\varepsilon)\pi_T - (R+\varepsilon)\mu \big]/m - \gamma_p (1-\varepsilon)\pi_T V$. The first-order conditions give

$$h_p^* = -\frac{\alpha d_3 + d_5}{2 d_4}, \qquad \alpha^* = \frac{d_3 d_5 - 2 d_2 d_4}{4 d_1 d_4 - d_3^2}, \qquad \hat{\alpha} = (\alpha^* \vee 0) \wedge \pi_0. \tag{16}$$

The minimal entropy is $\mathcal{H}(\mathbb{P}^* | \mathbb{P}^s) = \frac{T}{2} \big(\lambda - \rho \sigma_2 \gamma_s h_s / m \big)^2$, where $\lambda = (\mu_1 - r)/\sigma_1$ is the market price of risk of the stock, and the investor's problem is quadratic in h_s with

$$\hat{h}_s = \frac{\mathbb{E}[P_T] - F - k}{\bar{\gamma}_s V}, \qquad \bar{\gamma}_s = \gamma_s (1 - \rho^2), \tag{17}$$

where k is the covariance-hedging term. The authors print k as $\lambda \rho T \operatorname{Var}[P_T]$ in (5.4) and (A.17). The reviewer's own derivation is as follows. Let the investor choose stock volatility σ' with Sharpe ratio λ; then $\mathbb{E}[G] = \lambda \sigma' T$ and $\operatorname{Var}(h_s P_T + G) = h_s^2 V + \sigma'^2 T + 2\rho h_s \sqrt{V} \sigma' \sqrt{T}$, and the mean-variance objective $h_s(\mathbb{E}P_T - F) + \lambda \sigma' T - \frac{\gamma_s}{2} \operatorname{Var}$ is maximised at $\sigma' = \lambda/\gamma_s - \rho h_s \sqrt{V}/\sqrt{T}$. Substituting gives the value $h_s(\mathbb{E}P_T - F) - \frac{1}{2}\gamma_s (1-\rho^2) V h_s^2 - \lambda \rho h_s \sqrt{TV} + \lambda^2 T/(2\gamma_s)$, hence

$$k = \lambda \rho \sqrt{T V} = \lambda \rho \, \frac{\sigma_2 T}{m}. \tag{18}$$

CRITIQUE A term $\lambda \rho T \operatorname{Var}[P_T]$ has the dimension of price squared times time and cannot be subtracted from a price difference. Equation (18) has the dimension of price. It is possible that the square root was lost in the HTML conversion of the source; the reviewer uses (18) throughout, and the numerical results below reproduce the authors' qualitative statements with it.

3.6 Reconstruction of the equilibrium forward price

INFERENCE With no storage ($\hat{\alpha} = 0$), the producer's forward position is $h_p = (\mathbb{E}P_T - F)/(\gamma_p V) - \pi_T$ (Remark 5.2) and market clearing $h_p + h_s = 0$ gives

$$\mathbb{E}[P_T] - \hat{F} = \frac{\gamma_p \bar{\gamma}_s}{\gamma_p + \bar{\gamma}_s} V\Big(\pi_T + \frac{k}{\bar{\gamma}_s V}\Big). \tag{19}$$

The left-hand side is the forward premium in price units. The harmonic combination $\gamma_p\bar{\gamma}_s/(\gamma_p + \bar{\gamma}_s)$ is the risk aversion of the pooled risk-sharing pair; $V\pi_T$ is the risk-bearing demanded by the producer's unhedged output; and $k/\bar{\gamma}_s V$ is a hedging-pressure term that is positive when the demand shock is positively correlated with stocks and negative otherwise. Two predictions follow: the premium is increasing in γ_p and in γ_s, and it depends on ρ through $\bar{\gamma}_s$ (even in ρ) and through k (odd in ρ). The authors state that equilibrium quantities depend mainly on ρ^2; (19) shows that this holds only when $\lambda \approx 0$ or when the sign of ρ is ignored.

3.7 Convenience yield

PAPER The convenience yield y solves

$$F = P_0 \frac{1 + R}{1 - \varepsilon} - y P_0. \tag{20}$$

It is the proportional gap between the cost-of-carry forward price and the equilibrium forward price, per unit of spot price. In this static model it is not a flow of consumption benefit; it is a residual that absorbs the hedging pressure in F and the price effect of storage in P_0.

4 Data and Empirical Strategy

4.1 Data

The paper uses no data set. All empirical statements are drawn from the cited literature (index-fund holdings growing from 15 to over 200 billion dollars between 2003 and 2008; correlations between GSCI returns and the stock market ranging between -38% and $+40\%$ by period). The numerical figures use parameter values that are not listed in the text read, so they cannot be replicated from the paper alone. Table 1 lists what the model takes as inputs.

Table 1. Model inputs and the sources of variation (no estimation is performed in the paper)

Input	Role	Source of value
μ, m	Demand level and slope	Free parameters; not calibrated
π_0, π_T	Production at dates 0 and T	Free; the scenarios $\pi_T/\pi_0 = 0.3$ and 0.6 appear in Figure 5.3
γ_p, γ_s	CARA risk aversion	Free; varied across figures
$\rho, \sigma_1, \sigma_2, \lambda$	Stock and demand dynamics	Free; ρ is the main comparative-statics variable
$\eta_1, \eta_2, \lambda_J$	Jump sizes and intensity	Free; jump of the stock set to zero in Figure 5.5

Source: reviewer's summary of Sections 2, 4 and 5 of the paper (official content, restated).

4.2 Empirical specification and identification

There is no identification problem in the econometric sense because nothing is estimated. The analogue is the origin of comparative-statics variation: the model is solved at one set of parameters at a time and a single parameter is moved. The variation is therefore structural by construction, and the causal claims ("more investors raise the spot price") are statements about the model's solution, not about data. Representing a larger number of investors by a smaller γ_s relies on CARA aggregation (the paper cites Wilson, 1968): aggregate risk tolerance is the sum of individual tolerances. This is exact only with identical beliefs and exponential utility, and it excludes entry costs and heterogeneous constraints.

4.3 Data limitations and biases

- The model is calibrated to nothing, so none of the usual data biases apply, but the empirical support invoked in the introduction (comovement of fund holdings and prices) is correlational and the direction of causation is the very point in dispute in the empirical literature the paper cites.
- Parameter choices in Figures 5.1 to 5.7 are not reported in the text, which prevents replication and makes the qualitative monotonicities impossible to separate from parameter regions, an issue made concrete below.

5 Results

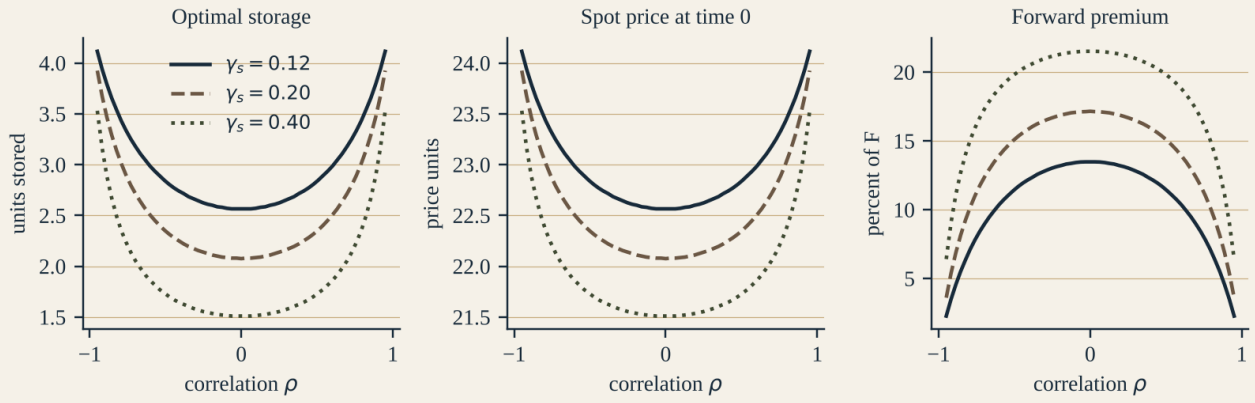
5.1 Existence and uniqueness

PAPER Theorem 3.9 states existence under conditions (EM), (COE), (USC), (NA) and a regularity condition (3.25). Proposition 4.12 states uniqueness of $\hat{F}$ in the Lévy setting. EVIDENCE These are mathematical results and are established by the proofs given, which this review followed step by step, subject to the concavity point above. Their economic content is limited: they say that a price exists, not that it is stable or reachable by any trading process.

5.2 Comparative statics in the Brownian case

EVIDENCE Because the authors' figures were not available as images, the reviewer re-implemented the Brownian equilibrium from (14) to (19) (storage by exact grid maximisation over α, forward price by root-finding on the clearing condition) with illustrative parameters of the reviewer's choosing: $\mu = 30$, $m = 1$, $\pi_0 = 10$, $\pi_T = 6$, $\varepsilon = R = 0.02$, $\sigma_2 = 2$, $T = 1$, $\gamma_p = 0.2$, $\lambda = 0$. These are generated, not the authors' numbers. Figure 1 and Table 2 report the outcome.

Figure 1. Equilibrium storage, spot price and forward premium as functions of the correlation ρ, for three investor risk aversions γ_s (generated, Brownian model)



Source: generated by the reviewer from the model of Section 5.1 of the paper; parameters: $\mu = 30$, $m = 1$, $\pi_0 = 10$, $\pi_T = 6$, $\varepsilon = R = 0.02$, $\sigma_2 = 2$, $T = 1$, $\gamma_p = 0.2$, $\lambda = 0$. Units are arbitrary price and quantity units; the premium is $(\mathbb{E}[P_T] - F)/F$ in percent. Not from the paper.

The figure shows that, for every γ_s, storage and the spot price increase with $|\rho|$ and the forward premium falls, with curves symmetric in ρ because $\lambda = 0$. A lower γ_s shifts storage and the spot price up and the premium down at every ρ, which is the equivalence between ρ^2 and γ_s stated by the authors. The effect is nonlinear and concentrated at high correlation, because $1 - \rho^2$ falls quickly only near $|\rho| = 1$.

Table 2. Re-computed Brownian equilibrium: spot price and forward premium (generated)

	Spot price $\widehat{P}_0$			Forward premium, percent		
γ_s	$\rho = 0$	0.5	0.9	$\rho = 0$	0.5	0.9
0.12	22.56	22.84	23.85	13.5	11.5	4.1
0.20	22.07	22.35	23.55	17.2	15.1	6.3
0.40	21.51	21.73	23.00	21.5	19.8	10.3

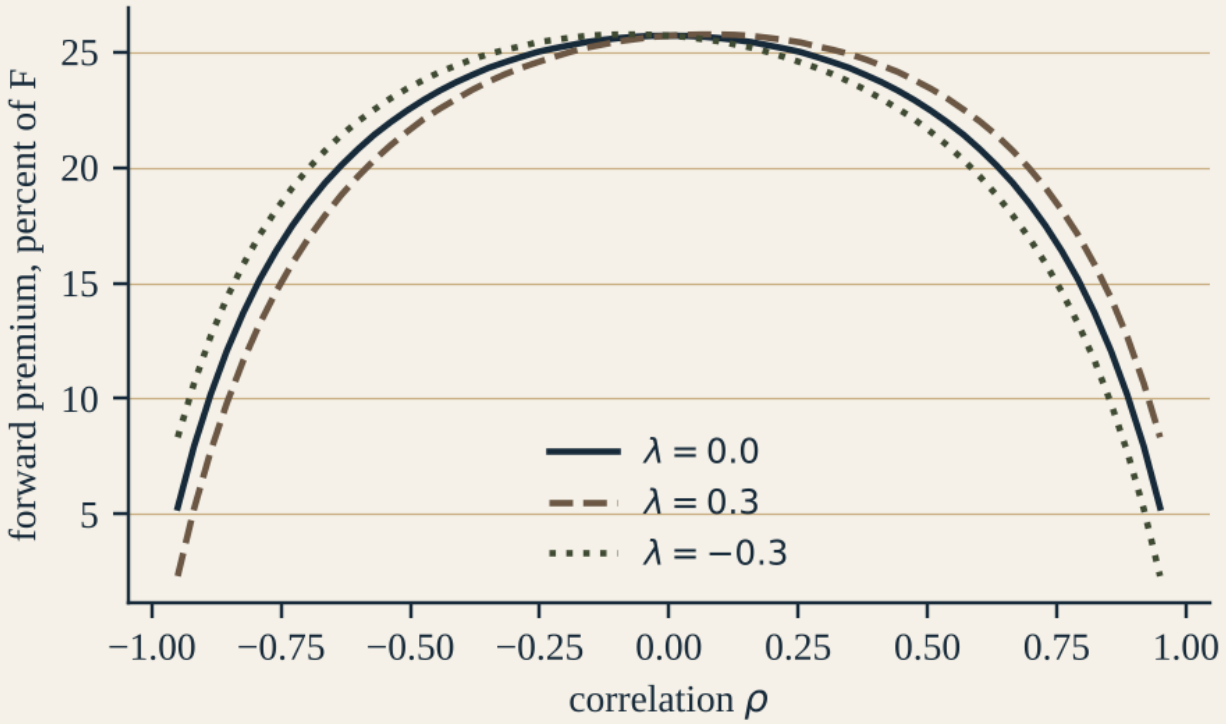
Source: generated by the reviewer; parameters as in Figure 1. Not official and not from the paper.

At $\gamma_s = 0.2$, moving from $\rho = 0$ to $\rho = 0.9$ lifts the spot price from 22.07 to 23.55, about 6.7% and about 0.74 of a standard deviation of P_T (the standard deviation is $\sqrt{V} = 2$), and cuts the premium from 17.2% to 6.3%. These magnitudes are a property of the arbitrary scale chosen; they illustrate direction and shape, not empirical size. The effect of the investor's risk aversion at $\rho = 0$ is the same order: doubling γ_s from 0.20 to 0.40 lowers the spot price by 2.5% and raises the premium by 4.3 points.

5.3 Effect of the stock-market price of risk

INFERENCE The paper's discussion is in terms of ρ^2. With $\lambda \neq 0$ the premium has an odd component. The generated calculation (Figure 2) shows that at $\lambda = 0.3$ the premium is lower than at $\lambda = 0$ for $\rho < 0$ and higher for $\rho > 0$: positively correlated demand shocks make the forward riskier for the investor's portfolio, while negatively correlated shocks make it a hedge for which the investor accepts a lower premium. In principle, forward prices above the expected spot price become possible for strongly negative ρ and large λ; the calculation does not reach that region at these parameters.

Figure 2. Forward premium as a function of correlation for three values of the stock-market Sharpe ratio λ (generated, Brownian model)



Source: generated by the reviewer; same parameters as Figure 1 with $\gamma_s = 0.2$, $\pi_T = 10$. The $\lambda \neq 0$ cases use $k = \lambda\rho\sqrt{TV}$ from the reviewer's derivation, Equation (18). Not from the paper.

5.4 Effect of the forward market on storage and price stability

PAPER Without a forward contract the producers' only motive to store is uneven production, $\alpha^* = -d_2/(2d_1)$ (Remark 5.3); with a forward contract storage is always weakly higher (Figure 5.3), the spot price is higher, and the expected future spot price lower. When π_T is small relative to π_0, the expected change $(\mathbb{E}\hat{P}_T - \hat{P}_0)/\hat{P}_0$ is closer to zero with the forward market (Figure 5.4). CRITIQUE "Stabilisation" here means a smaller expected price change in a model with one date; it is not a statement about price volatility, which in this model is fixed by the exogenous demand shock.

5.5 Jumps

PAPER With a common Poisson shock in X and Y (Equation (5.10)), the variance of P_T rises to $(\sigma_2^2 + \lambda\eta_2^2)T/m^2$, producers store more and sell more forwards, and the premium rises irrespective of the sign of the jump when the jump is unhedgeable ($\eta_1 = 0$). CRITIQUE The result is largely a variance effect: Remark 5.5 notes that an independent Brownian motion with the same variance gives the same first two moments of P_T and a higher premium. The jump example therefore adds fat tails but does not isolate any distinctive jump channel.

6 Robustness and Additional Tests

The paper reports no robustness tests in the empirical sense. Its additional analyses are structural variations, assessed in Table 3 in the same format (concern, test, result, assessment).

Table 3. Additional analyses in the paper and their assessment

Concern	Test reported	Result	Does it resolve the concern?
Results driven by Gaussian shocks	Poisson jump added to X and Y (Section 5.2)	Same signs of effects; higher premium with jumps	Partly: variance rather than jump structure explains it (Remark 5.5); one jump size and sign symmetric
Storage motive is only uneven production	Models with and without forward contract; $\pi_T = 0.3\pi_0$ and $0.6\pi_0$ (Figures 5.3, 5.4)	Forward market raises storage when output is uneven	Partly: demonstrates the mechanism but at unreported parameters
Hedge position from a previous cycle	Existing long forward b' added to wealth (Section 5.4)	Acts like higher π_T; less storage, higher convenience yield	Partly: static device; no dynamic optimisation of rolling
Dependence on risk-aversion parameters	Sweeps of γ_p, γ_s (Figures 5.1, 5.2, 5.6)	Monotone spot price, premium and yield	Partly: monotonicity depends on the concavity region of the producer's problem (below)
Dependence on correlation sign	Discussion of ρ^2	Effects mostly even in ρ	No: holds only when λ is near zero (Equation (19))

Source: reviewer's classification of the paper's Sections 5.2 to 5.4 and the associated figures (official content, assessed by the reviewer).

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Existence of an equilibrium forward price and of an optimal storage and hedging pair under CARA preferences, a convex trading set and the stated integrability conditions (Theorem 3.9), and uniqueness of the price when value functions are differentiable. The reduction of the investor's problem to a minimal entropy martingale measure computation under Lévy dynamics (Proposition 4.4) is a correct application of known results.

SUPPORTED BUT CONDITIONAL

The sign patterns of comparative statics (spot price increasing in γ_p and decreasing in γ_s; premium increasing in γ_s and decreasing in γ_p; spot price increasing in ρ^2). The reviewer's reproduction confirms them at one parameter point in the region where the producer's problem is jointly concave, with $\lambda = 0$.

SUGGESTIVE

That higher investor participation and higher stock correlation explain the observed comovement of fund holdings and commodity prices after 2003. The model delivers the same direction, but the data are not used, and other models with different channels deliver the same direction.

NOT ESTABLISHED

That financialisation raised spot prices in reality; that the convenience yield in the model is the convenience yield of the theory of storage; that price stabilisation in the model refers to volatility; that the results survive competitive (price-taking) producers, non-negativity of prices and quantities, or multiple periods.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Generality of the trading set $\mathcal{G}$ (transaction costs, constraints and dynamic strategies can be encoded), which makes the existence theorem portable.
- Inclusion of a correlated stock market and of jumps, which earlier equilibrium models of spot and forward commodity prices lack.
- Constructive proof that yields implementable fixed-point equations rather than a pure existence statement.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The sup-convolution and conjugate argument isolates the equilibrium price as a supergradient, a clean structure that extends to any pair of cash-invariant preferences, so the framework is reusable.
- The effective risk aversion $\bar{\gamma}_s = \gamma_s(1-\rho^2)$ is a transparent economic result: it quantifies how much an outside hedging opportunity adds to risk-bearing capacity and delivers a testable, parametric prediction linking the forward premium to a measurable correlation.
- The paper is honest about scope: the dynamic version is explicitly left for future research.

7.3 Weaknesses and critique

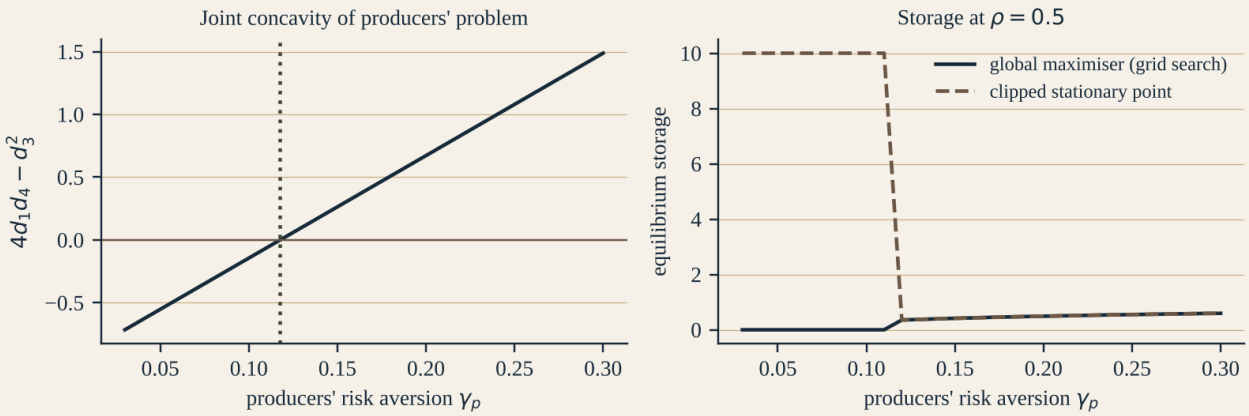
SINGLE MOST IMPORTANT WEAKNESS

Problem. The closed-form storage rule (16) is the stationary point of a quadratic programme in (α, h_p) that is clipped to $[0, \pi_0]$; it is the global maximiser only when the Hessian is negative definite, that is when $4d_1d_4 - d_3^2 > 0$. From the coefficients (15), $4d_1d_4 - d_3^2 = 2\gamma_p V(1+R)/m - (1-\varepsilon)^2/m^2$, which is positive if and only if

$$\gamma_p V m > \frac{(1-\varepsilon)^2}{2(1+R)}. \tag{21}$$

Below this threshold the profile of the objective after optimising over h_p is convex in α, the stationary point is a minimum, and the optimum lies at a corner of $[0, \pi_0]$. **Why it matters.** The authors' Propositions 5.1 and 5.4 are stated without this condition, and their comparative statics (storage increasing in γ_p, smooth responses in ρ) are derived from the interior formula. **Potential bias.** In the reviewer's computation with $V = 4$, $m = 1$, $\varepsilon = R = 0.02$ the threshold is $\gamma_p \approx 0.118$. For $\gamma_p = 0.10$ at $\rho = 0$ the exact optimum is $\alpha = 0$ while the clipped formula returns $\alpha = \pi_0 = 10$, the worst corner, with an implied forward premium of 107% against 20% at the true optimum (Figure 3). The error is not small and it reverses the sign of the storage response to γ_p below the threshold. **How it could be tested.** Verify (21) at the parameters of Figures 5.1 to 5.7, which the authors do not report, and recompute the equilibrium by brute-force maximisation of the producer's objective over α.

Figure 3. **Left: sign of $4d_1d_4 - d_3^2$ against the producers' risk aversion. Right: equilibrium storage at $\rho = 0.5$ from the exact maximiser and from the clipped stationary point (generated, Brownian model)**



Source: generated by the reviewer from the coefficients (A.5) of the paper; $\mu = 30$, $m = 1$, $\pi_0 = \pi_T = 10$, $\varepsilon = R = 0.02$, $\sigma_2 = 2$, $T = 1$, $\gamma_s = 0.2$, $\lambda = 0$. The dotted vertical line is the threshold in Equation (21). Not from the paper.

OTHER WEAKNESSES

- **Price-setting producers.** Producers internalise the effect of storage on P_0 and P_T, behaving like a cartel. The central result that more investor risk tolerance raises the spot price works through withholding supply. With price-taking producers the spot price effect of hedging would

operate only through expectations of P_T. Potential bias: overstated spot effects. Test: replace the representative producer with N Cournot or price-taking producers and let N grow.

- **Unbounded Gaussian demand shocks, linear demand and no non-negativity.** Prices, and quantities consumed, can be negative; the exponential moments assumed in (EM) make this inescapable for linear demand. Potential bias: the tails, which drive CARA risk premia, are mis-specified. Test: truncated demand or constant-elasticity demand and numerical equilibrium.
- **Static, two-date structure.** The convenience yield, the forward curve and the roll of positions cannot be studied, and "hedging by rolling" is invoked in Section 5.4 without a dynamic model. Test: a multi-period extension with state-dependent storage.
- **Investors as a single representative CARA agent.** Index investors are long-only and benchmarked, not CARA optimisers with unlimited short positions. In the paper's own words the correlation effect requires no short-selling constraint. Potential bias: the effective risk-aversion channel is overstated for constrained investors. Test: set $\mathcal{G}$ to a constrained set, which the paper allows but does not do.
- **Wealth effects absent.** CARA removes income effects: the risk premium does not depend on wealth. The effect of fund inflows on premia is therefore carried by γ_s only. Test: CRRA or habit preferences.
- **Dimensional inconsistency** in the printed covariance-hedging term, discussed above, which at best is a typographic artifact.

8 Contradictions and Edge Cases

CRITIQUE Several conditions make the result disappear or reverse.

Perfect correlation. As $|\rho| \to 1$, $\bar{\gamma}_s \to 0$ and the investor can replicate the demand shock with the stock index. From (17) the forward price is then pinned at $\widehat{F} = \mathbb{E}[P_T] - k$ with $k = \lambda\rho\sqrt{TV}$, independent of either risk aversion, and the investor's risk aversion no longer matters. The paper's headline result that investors' participation raises the spot price therefore weakens at high correlation: the investor's position is determined by the clearing condition, but its price is fixed by replication. This is consistent with Remark 4.11 (the strict concavity needed for a unique h_s fails when X is replicable) and is not highlighted.

Low producer risk aversion. Below the threshold (21) the model has bang-bang storage rather than smooth responses. In the example above, $\gamma_p = 0.10$ gives zero storage at both $\rho = 0$ and $\rho = 0.5$, so the spot price does not respond to correlation over that range and the main correlation channel vanishes there.

Equal production across dates. When $\pi_T \geq \pi_0$ the paper itself finds that storage without the forward market is zero and the forward market only creates storage through hedging pressure. Evaluated at the reviewer's parameters ($\pi_T = 10$, $\gamma_p = 0.2$), the equilibrium storage at $\rho = 0$ is 0.15 against $\pi_0 = 10$, about 1.5% of output: the spot price is lifted by only 0.15 above a level of 20, so the storage channel is economically small in that region.

Extreme values. As $\gamma_p \to \infty$ the producer hedges fully, $h_p \to -(\pi_T + \alpha(1-\varepsilon))$, the forward absorbs all risk and $\widehat{F} = \mathbb{E}[P_T] - \bar{\gamma}_s V(\pi_T + \alpha(1-\varepsilon)) - k$ so that the premium is proportional to $\bar{\gamma}_s V$ times total output, which can exceed the price itself: for $\bar{\gamma}_s V(\pi_T + \alpha(1-\varepsilon)) > \mathbb{E}[P_T] - k$ the forward price turns negative. Nothing in the model rules this out.

Sign of ρ and λ. For $\rho < 0$ and $\lambda > 0$, the term k is negative and enters (19) with the opposite sign: the forward can trade above the expected spot price even though producers want to hedge. This is the opposite of normal backwardation and it arises without any convenience yield, so the model cannot by itself discriminate between theories of storage and of hedging pressure.

Alternative mechanism. The spot-price effect of increased investor participation could equally arise from a shift in inventory policy of consumers or from learning; in the model only producers hold inventory. If consumers or intermediaries can also store, the spot price channel would depend on their responses.

9 Further Literature

Table 4. **Literature connections (from the paper's reference list; descriptions are the reviewer's)**

Paper	Shared idea	Relationship	Why it matters
Keynes (1930); Hicks (1939)	Normal backwardation: producers pay speculators a premium to hedge	Preceded; theoretical foundation of the premium	Provides the sign convention for $\mathbb{E}[P_T] - F$
Kaldor (1939); Working (1949); Brennan (1958)	Theory of storage and convenience yield	Preceded; second foundation	Defines the yield compared with in Equation (20)
Stoll (1979); Hirshleifer (1988, 1989)	Futures risk premia with trading costs and storage in mean-variance form	Developed similar ideas with Gaussian preferences	The Brownian case here nests them
Anderson and Danthine (1980, 1983)	Hedging and storage with exponential utility	Method precedent for the producer's problem	Source of the structure of $\bar{w}$
Routledge, Seppi and Spatt (2000)	Equilibrium forward curves with storage and a no-negative-inventory constraint	Different methodology; risk-neutral	Contains the dynamic and constrained features this paper lacks
Acharya, Lochstoer and Ramadorai (2013)	Limits to arbitrage and hedging with empirical test on oil and gas	Special case; supplies the empirical test the paper leaves open (Remark 5.6)	Gives a template for confronting the model with data
Gorton, Hayashi and Rouwenhorst (2012)	Fundamentals of commodity futures returns	Empirical counterpart; inventory-based risk premia	Tests the inventory channel in data
Basak and Pavlova (2016)	Financialisation with institutional investors and benchmarks in a dynamic equilibrium	Alternative explanation of the same facts	Delivers cross-commodity comovement through investor mechanisms not present here
Knittel and Pindyck (2013)	Speculation can raise spot prices only through higher inventories	Contradicts or conditions the spot-price channel	Same channel as this paper (storage), which makes the missing inventory evidence critical
Singleton (2014); Hamilton and Wu (2015); Henderson, Pearson and Wang (2015)	Empirical evidence on index flows and commodity prices	Test the direction of the claim	Evidence for and against the investor-participation channel

Paper	Shared idea	Relationship	Why it matters
Hubalek and Sgarra (2006); Fujiwara and Miyahara (2003)	Minimal entropy martingale measure for exponential Lévy models	Technical foundation of Proposition 4.4	Makes the semi-explicit solution possible
Cheridito, Horst, Kupper and Pirvu (2016); Filipović and Kupper (2008)	Equilibrium pricing with translation-invariant preferences	Same mathematical family	Source of the duality and sup-convolution methods

Source: reference list of the paper (official bibliographic details); the characterisation of each work's content is the reviewer's, based on the title and the paper's own description. The works other than the paper itself were not opened in this review; descriptions of Knittel and Pindyck (2013) and Basak and Pavlova (2016) rely on the paper's description and the reviewer's background knowledge and should be checked before use.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** Parameters behind the figures are not reported and the concavity region of the producer's problem is not stated. *Why it matters:* comparative statics are region-specific. *Existing limitation:* replication is impossible. *Way forward:* state condition (21) and publish parameters.
- **Gap:** No test of the predicted link between ρ and forward premia (deferred in Remark 5.6). *Why it matters:* it is the only falsifiable output. *Existing limitation:* correlation is time-varying and endogenous. *Way forward:* panel regression with rolling correlations and hedging-pressure measures.
- **Gap:** No analysis of welfare or of consumers' surplus. *Why it matters:* stabilisation claims need a welfare benchmark. *Existing limitation:* consumers are passive. *Way forward:* compute surplus with and without the forward market.

10.2 Literature-level gaps

- **Gap:** Whether investor flows move spot prices requires inventories and flows to be observed together. *Why it matters:* the channel in models of this type operates through inventories. *Existing limitation:* inventory data are partial and lagged. *Way forward:* commodities with good stock data, using flows as shocks.
- **Gap:** Dynamic equilibrium with storage, risk-averse hedgers and a stock market. *Why it matters:* term structure. *Existing limitation:* analytical intractability. *Way forward:* numerical dynamic programming.

10.3 Methodological gaps

- **Gap:** Equilibrium under position limits and benchmarking. *Why it matters:* index investors are constrained. *Existing limitation:* the set $\mathcal{G}$ is flexible in theory but only the unconstrained case is solved. *Way forward:* constrained duality, numerical solution.
- **Gap:** Non-Gaussian, bounded demand. *Why it matters:* tails drive CARA premia. *Existing limitation:* the (EM) condition. *Way forward:* truncated distributions and a numerical fixed point on the system (5.12) and (5.13).

11 Research Opportunities

Idea 1. Test of the correlation and hedging-pressure prediction on forward premia

RESEARCH QUESTION. Do commodity forward premia decrease with the squared rolling correlation of the commodity with equities, and does the odd component predicted by a nonzero stock-market Sharpe ratio appear in the sign of the correlation?

MOTIVATION. Equation (19) is the only closed-form prediction the paper offers for a cross-section of commodities, and it has two separable parts, even and odd in ρ.

HYPOTHESIS. Premium decreases in ρ^2 and is higher for positive than for negative ρ when the equity premium is positive.

ECONOMIC MECHANISM. Effective risk aversion $\gamma_s(1-\rho^2)$ plus a CAPM-type hedging term $\lambda\rho\sqrt{TV}$.

LITERATURE CONNECTION. Acharya, Lochstoer and Ramadorai (2013); Gorton, Hayashi and Rouwenhorst (2012); Henderson, Pearson and Wang (2015).

DATA. Futures settlement prices by contract maturity from exchange or vendor sources; CFTC Commitments of Traders for producer hedging positions; equity index returns. Availability to be confirmed; the CFTC series is public.

VARIABLES. Premium proxy: the difference between the futures price and the subsequent spot price at maturity; rolling correlation with the S&P 500; hedger net short ratio; realised variance.

METHOD. Panel regressions with commodity and time fixed effects, with ρ^2 and ρ entering separately and interacted with hedger positions.

IDENTIFICATION STRATEGY. Within-commodity variation in correlation around macro regimes; instrument correlation by lagged cross-asset shocks.

EXPECTED CONTRIBUTION. First direct test of the effective-risk-aversion channel.

MAIN RISK. Realised premia are noisy and correlation is estimated with error; the static model may not map to dynamic roll returns.

Idea 2. Equilibrium with constrained, long-only index investors

RESEARCH QUESTION. How do position limits and long-only benchmarking change the spot-price and forward-premium effects of investor inflows?

MOTIVATION. The correlation channel in the paper requires unrestricted short selling; real index flows are constrained.

HYPOTHESIS. With long-only constraints the effective risk aversion channel is weakened for $\rho < 0$ and the inflow-to-premium relation becomes asymmetric.

ECONOMIC MECHANISM. The trading set $\mathcal{G}$ becomes a cone with bounds; the investor's value function is no longer quadratic in h_s.

LITERATURE CONNECTION. Cheridito, Horst, Kupper and Pirvu (2016); Basak and Pavlova (2016).

DATA. Simulation first; calibration to index weights and positions from CFTC index trader reports.

VARIABLES. Constraint level, ρ, γ_s, aggregate inflow as a share of production.

METHOD. Numerical solution of the dual problem on a grid and root-finding for F.

IDENTIFICATION STRATEGY. Model-based; comparative statics only.

EXPECTED CONTRIBUTION. Shows which results of the paper survive realistic constraints.

MAIN RISK. Loss of closed forms and of uniqueness.

Idea 3. Price-taking versus price-setting producers

RESEARCH QUESTION. How much of the spot-price effect of investor participation is due to producers' market power?

MOTIVATION. The main spot result depends on producers internalising $dP_0/d\alpha = 1/m$.

HYPOTHESIS. With N price-taking producers the spot-price effect of γ_s shrinks at rate $1/N$.

ECONOMIC MECHANISM. Each producer ignores its price impact; storage responds only to expected price differences.

LITERATURE CONNECTION. Routledge, Seppi and Spatt (2000); Knittel and Pindyck (2013).

DATA. Simulation with the paper's quadratic structure.

VARIABLES. Number of producers, heterogeneity of γ_p, elasticity m.

METHOD. Nash and competitive equilibria of the quadratic programme; closed forms available in the Gaussian case.

IDENTIFICATION STRATEGY. Model-based.

EXPECTED CONTRIBUTION. Separates the risk-sharing channel from the market-power channel.

MAIN RISK. The result may be known in the industrial organisation literature.

Idea 4. Multi-period equilibrium with storage and the term structure

RESEARCH QUESTION. Does the effective-risk-aversion channel produce a hump-shaped convenience yield term structure when storage is dynamic?

MOTIVATION. Section 5.4 explicitly leaves the dynamic roll of hedges for future work.

HYPOTHESIS. A dynamic version produces a convenience yield that falls with maturity and with ρ^2 at short maturities.

ECONOMIC MECHANISM. Inventories as a state variable, CARA preferences and a stock factor correlated with demand shocks.

LITERATURE CONNECTION. Routledge, Seppi and Spatt (2000); Casassus, Collin-Dufresne and Routledge (2008).

DATA. Simulation; calibration to futures curves of crude oil or natural gas.

VARIABLES. Inventory, demand factor, equity factor, maturity.

METHOD. Backward induction on a state grid or deep-learning solution of the Euler system.

IDENTIFICATION STRATEGY. Model-based, with matching of curve slopes.

EXPECTED CONTRIBUTION. A tractable benchmark of financialisation in a dynamic storage model.

MAIN RISK. Computation and the non-negativity constraint on inventory.

Idea 5. Non-Gaussian bounded demand and skewed premia

RESEARCH QUESTION. How do skewness and truncation in demand shocks change premia compared with the Gaussian benchmark of the same variance?

MOTIVATION. Remark 5.5 shows that the jump example mostly reproduces a variance effect.

HYPOTHESIS. Premia depend on the skewness of X through the third cumulant of κ_X, and negative skew raises the premium paid by producers who are long the commodity.

ECONOMIC MECHANISM. κ_X enters the objective through (7); a third-order expansion gives a skewness correction proportional to $\gamma_p^2 \ell^3$.

LITERATURE CONNECTION. Carr, Geman, Madan and Yor (2002); Gorton and Rouwenhorst (2006).

DATA. Commodity option-implied skewness (vendor data) to calibrate tempered stable or variance-gamma shocks; availability to be confirmed.

VARIABLES. Variance, skewness, kurtosis of X, γ_p, γ_s.

METHOD. Numerical solution of (5.12) and (5.13), plus a cumulant expansion to isolate the skewness effect.

IDENTIFICATION STRATEGY. Fix the first two moments and vary the third.

EXPECTED CONTRIBUTION. Identifies a distinctive jump channel that the paper's example does not isolate.

MAIN RISK. Small effects relative to variance; calibration of skewness is noisy.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Premium test	4	4	4	4	5	5	26
2. Constrained investors	4	3	4	4	4	4	23
3. Price-taking producers	3	5	5	4	4	3	24
4. Dynamic term structure	5	2	3	3	5	5	23
5. Skewed demand	4	3	3	4	3	3	20

Source: reviewer's subjective scoring.

The top three are Ideas 1, 3 and 2 (totals 26, 24 and 23, Idea 4 tying Idea 2). Idea 1 is the only one that confronts the model with data and tests its most distinctive prediction. Idea 3 is the cheapest and clarifies which part of the headline result is risk-sharing and which is market power. Idea 2 is chosen over Idea 4 for tractability: it keeps the static closed-form skeleton while testing the paper's most restrictive assumption, unlimited short selling by the investor.

12 Learning Framework

12.1 Prerequisite concepts

- Exponential (CARA) utility, certainty equivalents and entropic risk measures.
- Convex conjugates, sup-convolution and supergradients (Rockafellar, 1997).
- Cumulant generating functions and Lévy processes; the Esscher transform.
- Theory of storage and normal backwardation; convenience yield; forward premium.
- Indifference pricing and minimal entropy martingale measures.

12.2 Core concepts learned

- A forward price is a supergradient of the pooled value function at zero net position.
- An outside hedge lowers effective risk aversion by a factor $1-\rho^2$.
- Storage is a price-setting choice that couples spot price, expected future price and risk exposure.
- Existence of an equilibrium does not imply concavity of the producer's problem or interior storage.

12.3 Important equations

Equation (4) links storage to both spot prices; (7) turns the CARA objective into a cumulant generating function; (9) defines the equilibrium price by sup-convolution; (12) reduces the stock-market problem to an entropy; (17) with $\bar{\gamma}_s = \gamma_s(1-\rho^2)$ gives the investor's demand; (19) gives the premium; (21) gives the concavity threshold.

12.4 Methods learned

Convex duality for equilibrium existence; change of measure to split a joint hedging problem; minimal entropy martingale measure for exponential utility; numerical equilibrium by clearing conditions; and, from the review, how to check second-order conditions of a clipped quadratic programme.

12.5 Questions to test understanding

COMPREHENSION

1. What does a producer gain and risk by storing one more unit, and how does this appear in P_0 and $\mathbb{E}[P_T]$?
2. Why does the forward market clear at a price in the supergradient of $u(\alpha, 0)$?
3. What is the effective risk aversion of the investor and why does correlation lower it?

ANALYSIS

1. Derive (17) with the term (18) from a mean-variance problem over h_s and a stock position.
2. Show that $4d_1d_4 - d_3^2 = 2\gamma_p V(1+R)/m - (1-\varepsilon)^2/m^2$.
3. Why is the Brownian CARA problem equivalent to mean-variance, and where does that equivalence fail with jumps?

RESEARCH

1. How would you test the odd-in-ρ component of the premium, and what would reject the model?
2. Which result of the paper survives with price-taking producers?
3. How would you extend the model so that the convenience yield is a state variable?

12.6 Papers to read next

Routledge, Seppi and Spatt (2000); Acharya, Lochstoer and Ramadorai (2013); Basak and Pavlova (2016); Knittel and Pindyck (2013); Cheridito, Horst, Kupper and Pirvu (2016).

13 Conclusion

13.1 One-sentence thesis

Under CARA preferences, a risk-averse storing producer and a risk-averse investor with access to a correlated stock market have an equilibrium spot and forward price whose investor-side risk aversion is scaled down by $1-\rho^2$, which links correlation, spot prices and forward premia.

13.2 Three most important insights

- Correlation with the stock market works as a reduction of the investor's risk aversion, $\bar{\gamma}_s = \gamma_s(1-\rho^2)$.
- The spot-price effect of hedging comes from producers withholding supply, so it is a market-power effect as much as a risk-sharing one.
- Existence is generic; the shape of the equilibrium depends on second-order conditions that the paper does not state.

13.3 Strongest evidence

The existence theorem and the closed-form Brownian solution, which the reviewer reproduced numerically in the concave region.

13.4 Biggest weakness

The storage rule (16) is valid only if $\gamma_p Vm > (1-\varepsilon)^2/(2(1+R))$; outside that region the formula selects the worst corner.

13.5 Most important research gap

No empirical test of the correlation and premium prediction, and no dynamic version.

13.6 Best research opportunity

A cross-commodity panel test of the premium formula (Idea 1).

13.7 Three papers to read next

Routledge, Seppi and Spatt (2000); Acharya, Lochstoer and Ramadorai (2013); Basak and Pavlova (2016).

14 References

Acharya, V. V., Lochstoer, L. A., and Ramadorai, T. (2013). Limits to arbitrage and hedging: Evidence from commodity markets. *Journal of Financial Economics*, 109, 441 to 465.

Anderson, R. W., and Danthine, J.-P. (1980). Hedging and joint production: Theory and illustrations. *Journal of Finance*, 35, 487 to 498.

Anderson, R. W., and Danthine, J.-P. (1983). Hedger diversity in futures markets. *Economic Journal*, 93, 370 to 389.

Anthropelos, M., Kupper, M., and Papapantoleon, A. (2015). An equilibrium model for spot and forward prices of commodities. arXiv:1502.00674, version 3, 21 January 2017.

Baker, S. D. (2014). The financialization of storable commodities. Preprint, SSRN 2348333.

Basak, S., and Pavlova, A. (2016). A model of financialization of commodities. *Journal of Finance*, forthcoming at the time of the paper.

Brennan, M. (1958). The supply of storage. *American Economic Review*, 48, 50 to 72.

Carr, P., Geman, H., Madan, D. B., and Yor, M. (2002). The fine structure of asset returns: An empirical investigation. *Journal of Business*, 75, 305 to 332.

Casassus, J., Collin-Dufresne, P., and Routledge, B. (2008). Equilibrium commodity prices with irreversible investment and non-linear technologies. NBER Working Paper 11864.

Cheridito, P., Horst, U., Kupper, M., and Pirvu, T. (2016). Equilibrium pricing in incomplete markets under translation invariant preferences. *Mathematics of Operations Research*, 41, 174 to 195.

Ekeland, I., Lautier, D., and Villeneuve, B. (2014). Speculation in commodity futures markets: A simple equilibrium model. Working paper, Université Paris-Dauphine.

Filipović, D., and Kupper, M. (2008). Equilibrium prices for monetary utility functions. *International Journal of Theoretical and Applied Finance*, 11, 325 to 343.

Fujiwara, T., and Miyahara, Y. (2003). The minimal entropy martingale measures for geometric Lévy processes. *Finance and Stochastics*, 7, 509 to 531.

Gorton, G. B., and Rouwenhorst, K. G. (2006). Facts and fantasies about commodity futures. *Financial Analysts Journal*, 62, 47 to 68.

Gorton, G. B., Hayashi, F., and Rouwenhorst, K. G. (2012). The fundamentals of commodity futures returns. *Review of Finance*, 17, 35 to 105.

Hamilton, J. D., and Wu, J. C. (2015). Effects of index-fund investing on commodity futures prices. *International Economic Review*, 56, 187 to 205.

Henderson, B. J., Pearson, N. D., and Wang, L. (2015). New evidence on the financialization of commodity markets. *Review of Financial Studies*, 28, 1285 to 1311.

Hicks, J. R. (1939). *Value and Capital*. Oxford University Press.

Hirshleifer, D. (1988). Residual risk, trading costs, and commodity futures risk premiums. *Review of Financial Studies*, 1, 173 to 193.

Hirshleifer, D. (1989). Futures trading, storage and the division of risk: A multiperiod analysis. *Economic Journal*, 99, 700 to 719.

Hubalek, F., and Sgarra, C. (2006). Esscher transforms and the minimal entropy martingale measure for exponential Lévy models. *Quantitative Finance*, 6, 125 to 145.

Kaldor, N. (1939). Speculation and economic stability. *Review of Economic Studies*, 7, 1 to 27.

Keynes, J. M. (1930). *A Treatise on Money*, Volume 2. Macmillan, London.

Knittel, C., and Pindyck, R. (2013). The simple economics of commodity price speculation. NBER Working Paper 18951.

Pirrong, C. (2011). *Commodity Price Dynamics: A Structural Approach*. Cambridge University Press.

Rockafellar, R. T. (1997). *Convex Analysis*. Princeton University Press.

Routledge, B. R., Seppi, D. J., and Spatt, C. S. (2000). Equilibrium forward curves for commodities. *Journal of Finance*, 55, 1297 to 1338.

Singleton, K. J. (2014). Investor flows and the 2008 boom/bust in oil prices. *Management Science*, 60, 300 to 318.

Stoll, H. (1979). Commodity futures and spot price determination and hedging in capital market equilibrium. *Journal of Financial and Quantitative Analysis*, 14, 873 to 894.

Wilson, R. (1968). The theory of syndicates. *Econometrica*, 36, 119 to 132.

Working, H. (1949). The theory of price of storage. *American Economic Review*, 39, 1254 to 1262.